

IPSec Site-to-Site Route-Based con IKEv2

Topología

R2

- WAN: 200.1.15.2/24
- LAN: 10.15.99.1/24
- Tunnel10: 172.16.1.1/30

R3

- WAN: 200.1.99.2/24
- LAN: 192.168.99.1/24
- Tunnel10: 172.16.1.2/30

Configuración de R1 (ISP)

```
enable
configure terminal

interface e0/0
 ip address 200.1.15.1 255.255.255.0
 no shutdown

interface e0/1
 ip address 200.1.99.1 255.255.255.0
 no shutdown

end
write memory
```

Configuración de R2 (IKEv2)

```
enable
configure terminal

hostname R2

interface e0/0
 ip address 200.1.15.2 255.255.255.0
 no shutdown
```

```

interface e0/1
  ip address 10.15.99.1 255.255.255.0
  no shutdown

ip route 0.0.0.0 0.0.0.0 200.1.15.1

crypto ikev2 proposal IKEV2-PROP
  encryption aes-cbc-256
  integrity sha256
  group 14

crypto ikev2 policy IKEV2-POLICY
  proposal IKEV2-PROP

crypto ikev2 keyring VPN-KEYRING
  peer R3
    address 200.1.99.2
    pre-shared-key cisco

crypto ikev2 profile IKEV2-PROFILE
  match identity remote address 200.1.99.2 255.255.255.255
  authentication local pre-share
  authentication remote pre-share
  keyring local VPN-KEYRING

crypto ipsec transform-set VPN-SET esp-aes 256 esp-sha256-hmac
  mode tunnel

crypto ipsec profile VPN-PROFILE
  set transform-set VPN-SET
  set ikev2-profile IKEV2-PROFILE

interface Tunnel10
  ip address 172.16.1.1 255.255.255.252
  tunnel source 200.1.15.2
  tunnel destination 200.1.99.2
  tunnel mode ipsec ipv4
  tunnel protection ipsec profile VPN-PROFILE

ip route 192.168.99.0 255.255.255.0 Tunnel10

end
write memory

```

Configuración de R3 (IKEv2)

```

enable
configure terminal

```

```

hostname R3

interface e0/0
 ip address 200.1.99.2 255.255.255.0
 no shutdown

interface e0/1
 ip address 192.168.99.1 255.255.255.0
 no shutdown

ip route 0.0.0.0 0.0.0.0 200.1.99.1

crypto ikev2 proposal IKEV2-PROP
 encryption aes-cbc-256
 integrity sha256
 group 14

crypto ikev2 policy IKEV2-POLICY
 proposal IKEV2-PROP

crypto ikev2 keyring VPN-KEYRING
 peer R2
  address 200.1.15.2
  pre-shared-key cisco

crypto ikev2 profile IKEV2-PROFILE
 match identity remote address 200.1.15.2 255.255.255.255
 authentication local pre-share
 authentication remote pre-share
 keyring local VPN-KEYRING

crypto ipsec transform-set VPN-SET esp-aes 256 esp-sha256-hmac
 mode tunnel

crypto ipsec profile VPN-PROFILE
 set transform-set VPN-SET
 set ikev2-profile IKEV2-PROFILE

interface Tunnel10
 ip address 172.16.1.2 255.255.255.252
 tunnel source 200.1.99.2
 tunnel destination 200.1.15.2
 tunnel mode ipsec ipv4
 tunnel protection ipsec profile VPN-PROFILE

ip route 10.15.99.0 255.255.255.0 Tunnel10

end
write memory

```

Verificación

Limpiar asociaciones

```
clear crypto ikev2 sa  
clear crypto ipsec sa
```

Verificar IKEv2

```
show crypto ikev2 sa
```

Estado esperado:

```
READY
```

Verificar IPSec

```
show crypto ipsec sa
```

Verificar túnel

```
show interface tunnel10
```

Resultado esperado:

```
Tunnel10 is up  
line protocol is up
```

Pruebas de conectividad

```
ping 172.16.1.2  
ping 192.168.99.1  
ping 192.168.99.2
```